

INDIAN INSTITUTE OF TECHNOLOGY PALAKKAD

ENGINEERING DESIGN PROJECT

ARDUINO CALCULATOR

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IIT PALAKKAD

CONTENTS:

1. OVERVIEW
2. OBJECTIVE
3. COMPONENTS USED
4. CIRCUIT DESIGN
5. PROCESS OF THE OPERATION
6. PRACTICAL USAGE
7. BUDGET

1. OVERVIEW

The project involves building the calculator using Arduino this helps us gain a better understanding of the workings of a calculator and helps us to understand the concepts you can implement even scientific functions with Arduino's built-in functions.

2. OBJECTIVE

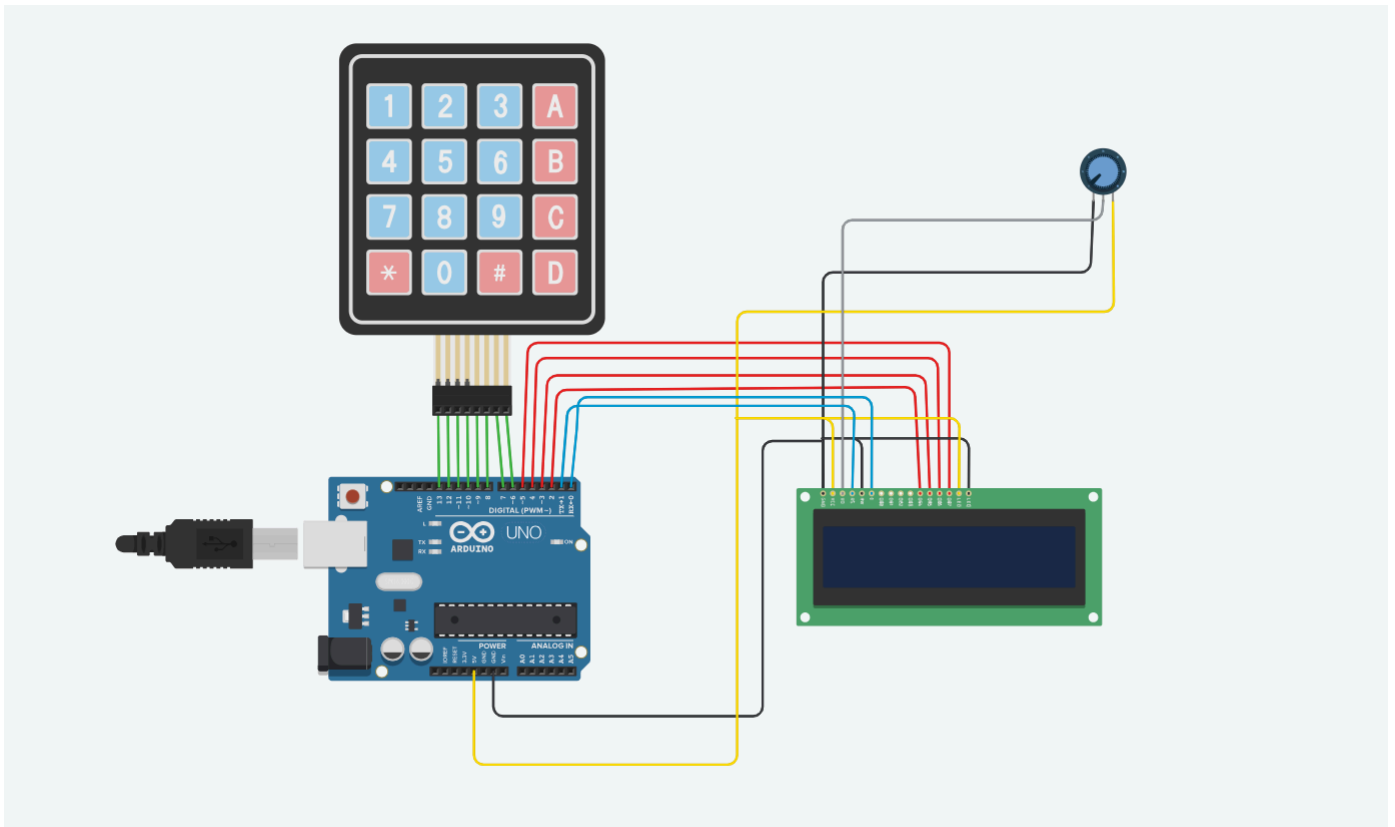
1. A working calculator that takes input using the 4x4 keypad
2. Input getting read by the Arduino and able to perform addition, multiplication, subtractions and division
3. Getting the output using the LCD display.

3. COMPONENTS USED

1. Arduino UNO
2. 16X2 LCD DISPLAY
3. 4X4 KEYPAD
4. BREADBOARD
5. JUMPER CABLES
6. POTENTIOMETER

4. CIRCUIT DESIGN

THE FOLLOWING IS THE THE CIRCUIT OF THE PROJECT



5. PROCESS OF OPERATION

ARDUINO UNO R3: THE MICRO CONTROLLER WILL ACT AS THE CALCULATOR'S BRAIN AND HELP US TO CONVERT THE GIVEN INPUT TO THE SOLUTION WE NEED.

4X4 KEYPAD: HELPS US TO GIVE INPUT ON THE TO THE ARDUINO

16X2 LCD DISPLAY: DISPLAY SHOWS US THE GIVEN INPUT AND ACQUIRED OUTPUT

BREADBOARD: HELPS US TO CONNECT COMPONENTS WITHOUT WELDING

6. PRACTICAL USAGE

HELPS US TO CALCULATE LARGE CALCULATIONS QUICKLY

7.BUDGET

COMPONENTS	QUANTITY	ESTIMATED COST(INR)
ARDUINO UNO	1	1200
4X4 KEYPAD	1	400
16X2 LCD	1	800
BREADBOARD	1	80
JUMPER CABLES	-	150
POTENTIOMETER	1	100

*TOTAL BUDGET MAY VARY